

Name: Daffa Gabillah Maulana Gapur

HVAC Engineer | Ship System Engineer | Production Planner

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1. PORTFOLIO: <https://github.com/gabillah>

2. LANGUAGE CERTIFICATION

- (2026). TOEIC. Score: 665/990. Organizer: Politeknik Perkapalan Negeri Surabaya.

3. EDUCATION BACKGROUND

- Institution: Politeknik Perkapalan Negeri Surabaya.

Study program: D4 Teknik Permesinan Kapal.

Period: August 2022 – July 2026.

Location: Surabaya, Indonesia.

Thesis: Analysis of Engine-Propeller Matching for a 160 TEUs Container Ship using B-Series and AU-Series Propellers.

4. WORK EXPERIENCE

- Job position: Student Intern at PT PAL (Persero) 

Division: Design Division, Machinery Outfitting Department.

Period: 21st February 2025 – 7th July 2025.

Type of business: Shipbuilding.

Location: Surabaya, Indonesia.

Responsibilities:

- Designed and engineered ship piping systems in compliance with Lloyd's Register requirements, covering (2D & 3D) layout planning, firefighting, ballast, bilge, sewage treatment, fresh water and feed water cooling systems, fuel systems (MDO and HFO), compressed air, HVAC, refrigeration, and electrical systems. Achieved 15% reduction in total pipe length through optimized routing and sizing of all systems. Tools: manual calculations in spreadsheet; P&ID and technical drawings in AutoCAD; 3D machinery models in Rhinoceros; 3D piping in Revit; CFD validation in Pipe Flow Expert; Reporting using LaTeX (TeXworks, MiKTeX, XeLaTeX + BibTeX).

5. PROJECT EXPERIENCE

- (2026). Job-shop type manufacturing production planning. Linear programming optimization of operational/work-center/station/labor costs and queue/lead time. Tools: Odoo for production planning data; Python (Jupyter/Anaconda) for optimization modeling; LaTeX (TeXworks, MiKTeX, XeLaTeX + BibTeX) for reporting.
- (2025). Overhauled ship diesel engine. Tools: Performed measurements using a caliper, micrometer screw gauge, and dial bore gauge; Reporting using LaTeX (TeXworks, MiKTeX, XeLaTeX + BibTeX).
- (2025). Diesel fuel injection pressure testing. Adjust diesel fuel injection pressure according to product guide.
- (2025). Conducted ship shaft alignment. Tools: Performed measurements using a dial gauge.
- (2025). Designed an HVAC round ducting system with a 17,000 CFM capacity for the accommodation building of a 21,000 DWT oil tanker, in accordance with ASHRAE, CARRIER (static regain), and SNAME standards. Tools: Calculations performed using spreadsheet; Technical drawings produced using AutoCAD; Validated using Pipe Flow Expert software; Reporting using LaTeX (TeXworks, MiKTeX, XeLaTeX + BibTeX).
- (2025). Designed the propulsion and shaft system arrangement and performed Engine-Propeller Matching with a B-Series propeller for a 21,000 DWT oil tanker. Tools: Calculations performed using spreadsheet; Technical drawings using AutoCAD.
- (2024). Performed steel welding operations using SMAW and aluminium welding operations using OAW.
- (2024). Carried out vacuum processing and R-22 refrigerant charging on HVAC systems.
- (2023). Designed general arrangement and engine room arrangement drawings for a 21,000 DWT oil tanker. Tools: Calculations performed using spreadsheet; Technical drawing using AutoCAD.
- (2023). Designed ship lines plan for a 21,000 DWT oil tanker. Tools: Calculations performed using spreadsheet; Technical drawing using AutoCAD.
- (2022). Conducted non-destructive testing (NDT) (including magnetic particle testing, liquid penetrant testing, and ultrasonic testing) and destructive testing (DT) (including tensile testing, impact testing, and hardness testing) on welded alloy materials.
- (2022). Lathe, milling, and drilling machine operator for pump shaft components.

6. VOLUNTEER EXPERIENCE

- Job position: Staff Member at UKM Penalaran PPNS.

Division: PKM Division.

Period: September 2022 – December 2024.

Type of organization: Educational organization.

Location: Surabaya, Indonesia.

Responsibilities:

- Served as Head of Events for Internal Monitoring and Evaluation, PKP2 Simulation, and PKP2:
 - Drafted proposals, organized and executed events, and compiled activity reports.
 - Achieved 4 out of 5 teams advancing to the national finals. 1 team was unqualified as its members had already graduated prior to the official qualification.
- Served as Staff Member for the PIMPUS event:
 - Reviewed 15 PKM proposals submitted by PPNS students.
 - Prepared event equipment.
- Served as Staff Member for the PIMNAS preparation team:
 - Prepared event equipment.

- Job position: Staff Member at Marine Engineering Student Association.

Division: Communication and Information Division.

Period: January 2023 – February 2024.

Type of organization: Educational organization.

Location: Surabaya, Indonesia.

Responsibilities:

- Creating posts for the Marine Engineering department website
- Creating posters using Photoshop.

- Job position: Graphic Design Volunteer at MILOO Project PT. Suryakata Teknologi Indonesia.

Division: Design.

Period: April 2021 – June 2021.

Type of organization: Educational organization.

Location: Surabaya, Indonesia.

Responsibilities:

- a. Creating poster designs for talkshow events, using Figma, Photoshop, and Illustrator.
4. Job position: Coordinator of Graphic Design team (Volunteer) at Sense 2017–2018 (art festival organized by OSIS/MPK SMAN 6 Surabaya).
Division: Design.
Period: August 2017 – May 2018.
Type of organization: Art festival.
Location: Surabaya, Indonesia.
Responsibilities:
 - a. Creating Instagram feeds.
 - b. Making posters.
 - c. Documenting the event using a DSLR camera.
5. Job position: Staff of Graphic Design team (Volunteer) at Championsix 2016–2017 (sports festival organized by OSIS/MPK SMAN 6 Surabaya).
Division: Design.
Period: August 2016 – May 2017.
Type of organization: Sport festival.
Location: Surabaya, Indonesia.
Responsibilities:
 - a. Creating Instagram feeds.
 - b. Making posters.
 - c. Documenting the event using a DSLR camera.